

Guidelines for Reporting About Network Data (GRAND)

These guidelines for reporting about network data describe the minimum information that should always be reported about a network dataset in a manuscript. They should be used to describe the final network data that are analyzed, following any transformations. In most cases, additional information about the network will also be necessary that is specific to the context, research question, or analysis.

- Underlined terms are defined in the glossary. The glossary is provided to ensure clarity in the guidelines, however GRAND remains agnostic about terminology.
- **Bold** terms note guidelines for specific types of networks. Further adaptation may be necessary for special cases, or for types of networks not yet covered by GRAND.
- *Italicized* guidelines are optional, but describe additional information that may often be useful to report, particularly if the network is not described in more detail elsewhere.

1.1 Description: Network

The items in this section are intended to provide a description of the network dataset as a whole.

Item	What to report
Type	Report the type of network: <u>Whole</u> , <u>Two-mode</u> , <u>Hypergraph</u> , <u>Multiplex</u> , <u>Discrete Dynamic</u> , <u>Nested</u> , <u>Ego</u>
Source	Report the source of the network data: <u>Empirical</u> , <u>Generated</u>
Count	Multiplex: Report the number of relationships. Discrete Dynamic: Report the number of waves. Nested: Report the number of networks. Ego: Report the number of ego networks. If count < 5, report the exact values of items below for each network, relationship, or wave. If count ≥ 5, report the range.
System	Report the system, location, and date that the network represents, at an ethically appropriate level of detail (e.g., an American high school in 2005). Ego: Report the population of <u>egos</u>, location, and dates that the ego networks represent, at a ethically appropriate level of detail (e.g., Immigrants in New York in 2005).
Data	Report how to obtain the data, or why the data are not available.
Citation	<i>Provide a citation to a source where the data is more fully described, or that credits those who originally collected/generated the data.</i>

1.2 Description: Nodes

The items in this section are intended to describe the nodes in the network.

Item	What to report
Meaning	Report what the <u>nodes</u> are intended to represent using a word or phrase (e.g., students). Two-mode: Report what each type of node is intended to represent using a word or phrase (e.g., plants and pollinators). Ego: Report what the <u>alter</u> nodes are intended to represent using a word or phrase (e.g., close friends).
Count	Report the number of nodes. Two-mode: Report the number of each type of node. Discrete Dynamic: Report (the range of) the number of nodes present in each wave. Nested: Report (the range of) the number of nodes in each network. Ego: Report (the range of) the number of alters in each ego network.
Attribute	If the nodes have non-structural attributes (e.g., demographic characteristics), report which node attributes are available. <i>Describe how the attributes were measured, and provide sample descriptive statistics for each.</i>

1.3 Description: Edges

The items in this section are intended to describe the edges in the network.

Item	What to report
Direction	Report whether the <u>edges</u> are undirected or directed.
Weight	Report whether the edges are unweighted, weighted, or signed.
Structural	If the network contains <u>structural edges</u> , report their presence and meaning.
Meaning	Report what the edges are intended to represent using a word or phrase (e.g., friendship). Multiplex: Report what each type of edge is intended to represent. Directed: Report the meaning of the edge direction (e.g., from origin to destination). Signed: Report what the positive edges are intended to represent, and what the negative edges are intended to represent, using words or phrases (e.g., collaboration and opposition). Weighted: Report the meaning (e.g., frequency, similarity, intensity), level of measurement (e.g., ordinal, continuous), and range of weights.
Count	Report the number of edges (ignore weights, count directed edges separately). Multiplex: Report (the range of) the number of edges of each type. Signed: Report the number of positive edges and the number of negative edges. Discrete Dynamic, Nested, Ego: Not applicable.

2.1 Data: Procedure

The items in this section are intended to explain how the network data was obtained.

Item	What to report
Method	Empirical: Report the method used to collect the data using a word or phrase (e.g., survey, interview, observation, archival). <i>Provide a detailed description of the data collection methods, including (as applicable): how the network was bounded, how participants were recruited, the modality and procedures of data collection, and the exact wording of the question(s) used to collect data.</i> Generated: Report the name and citation of the data generating model (e.g., <code>sample_smallworld()</code> in <code>igraph v2.1.4</code>) and the values of the function parameters. <i>Provide a detailed description of the data generating model, including its theoretical foundation and the role of each parameter.</i>
Ethics	Empirical: Report whether the data collection received institutional or ethical approval, and if so, the approving entity and approval date. <i>Provide a detailed description of any approvals, procedures for obtaining informed consent, potential risks, and data security practices such as deidentification.</i>

2.2 Data: Error

The items in this section are intended to describe potential sources of error, bias, or distortion that arise from the data collection or generation process itself.

Item	What to report
Degree	If a node's minimum or maximum degree is constrained by the data collection or generation design, report the restriction on degree (e.g., respondents could name up to 5 contacts).
Nodes	If the network does not have a well-defined <u>boundary</u>, report that it is unbounded. If the network has a well-defined boundary, report the percent or number of missing nodes. Two-mode: If the network has a well-defined boundary, report the percent or number of each type of missing node. Nested: If the network has a well-defined boundary, report (the range of) the percent or number of missing nodes in each network. Ego: Not applicable.

2.3 Data: Transformation

The items in this section are intended to describe transformations that were performed on raw data to obtain the data that are analyzed.

Item	What to report
Symmetrizing	If directed edges were transformed into undirected edges, report the transformation method (e.g., confirmed edges, unconfirmed edges), and the fraction of connected dyads that had non-reciprocal edges before the transformation. <i>Provide a detailed description of the transformation method.</i>
Binarizing	If weighted edges were transformed into unweighted edges, report the transformation method (e.g., threshold, backbone model), and the fraction of edges that were deleted as a result of the transformation. <i>Provide a detailed description of the transformation method.</i>
Projecting	If the network was generated from two-mode or non-network data such that edges represent similarities, co-occurrences, or statistical associations, report the transformation method (e.g., projection, gaussian graphical model). <i>Provide a detailed description of the transformation method.</i>
Excluding	If any nodes have been excluded from the network (e.g., <u>isolates</u> , smaller components), report the exclusion criteria and fraction of nodes that were removed. <i>Provide a detailed rationale for the exclusion of these nodes, and the analytic implications of their exclusion.</i>
Imputing	If any missing nodes or edges have been imputed, report the imputation method, and the fraction of nodes or edges in the network that were imputed. <i>Provide a detailed description of the imputation method.</i>

3 Structure

The items in this section report basic structural properties of the network. It is intentionally brief because other relevant structural properties depend on the research question or analysis.

Item	What to report
Connected	Report whether the network is <u>connected</u> or <u>unconnected</u>. Directed: Report whether the network is strongly connected, weakly connected, or unconnected. Multiplex: Report the number of types of edge for which the network is connected. Discrete Dynamic: Report the number of waves in which the network is connected. Nested: Report the number of networks that are connected. Ego: Not applicable.
Connectivity	Report either the <u>edge density</u> or <u>mean degree</u>, ignoring edge weights if present. Two-mode: Report the mean degree for each type of node. Hypergraph: Report the mean degree for nodes and the mean size for edges. Multiplex: Report (the range of) the edge density or mean degree for each type of edge. Discrete Dynamic: Report (the range of) the edge density or mean degree in each wave. Nested: Report (the range of) the edge density or mean degree for each network. Ego: Report (the range of) the edge density for each ego network, and whether these values were computed with or without ego. Signed: Report edge density or mean degree separately for positive and negative edges.

Glossary: Network Types

Multiple types of networks exist, and in some cases require unique reporting guidelines. There is no widely-accepted set of terms for different types of networks. For clarity, this glossary defines the terms used in GRAND. GRAND distinguishes mutually exclusive types of networks in terms of (1) the number of different entities that nodes represent, (2) the number of different relationships that edges represent, (3) what an edge can connect, and (4) the count of networks in a single dataset.

Note: GRAND does not use the term “multi-layer”. Instead, multi-layer networks are classified by what the ‘layers’ represent (e.g., types of relationships → a multiplex network; waves in time → a discrete dynamic network).

Type	Entities	Relationships	Connect	Count	Possible Synonyms	Example: A network describing...
Whole	one	one	dyads	one	complete, one-mode, sociocentric, unipartite	Students connected by friendship
Two-mode	two	one	dyads of different entities	one	bipartite	Pollinators connected to the types of plants they pollinate
Hypergraph	one	one	2+ nodes	one	higher order	All authors on a given paper connected by a single edge
Multiplex	one	many	dyads	each relationship	–	Students connected by friendship; Students connected by bullying
Discrete Dynamic	one	one	dyads	each wave	temporal	Students connected by friendship in January; Students connected by friendship in February
Nested	one	one	dyads	each network representing a common type of system	–	Students in classroom A connected by friendship; Students in a classroom B connected by friendship
Ego	one	one	dyads	each ego	personal, egocentric	A focal person and their sources of social support connected by support provision
<i>For inclusion in future versions of GRAND</i>						
Multimode	many	many	dyads of different entities	one	–	Pollinators connected to islands where they live, plants connected to islands where they grow, birds connected to plants they pollinate
Multilevel	many	many	dyads	one	meta-network, network of networks	Students connected by friendship, teachers connected by advice seeking, and students connected to teachers by classroom membership
Continuous Dynamic	one	one	dyads	–	temporal, relational event	Students connected by speaking interactions that have specific start and stop times

Glossary: Terminology

Multiple terms often exist for a single network concept. To ensure clarity, this glossary defines the terms used in the reporting guidelines. However, GRAND remains agnostic about which term to use when synonyms exist.

Term (<i>possible synonyms</i>)	Definition
Alter	In an ego network, the alters are the non-ego nodes in the network. Typically they are the individuals identified by ego in response to a name generator. By definition, they are connected to ego, but may or may not be connected to each other.
Boundary	The boundary of a network defines which nodes are part of the system the network is intended to represent, and which are not. The number of nodes missing from a network is only known when a network's boundary is known.
Connected	A network is connected if there exists a path between every pair of nodes, and otherwise is unconnected. A directed network is weakly-connected if there exists a path between every pair of nodes <i>when ignoring edge direction</i> , and is strongly connected if such paths exist <i>when obeying edge direction</i> .
Edge (<i>link, tie, arc, connection, relationship</i>)	The relationships that connect nodes in a network, and whose arrangement gives a network its structure.
Edge density (<i>Density</i>)	Number of present edges divided by the number of possible edges, counting each present/possible directed edge separately.
Ego	In an ego network, ego is the focal respondent who provides the data and to whom all other network members are connected.
Empirical	Data that are derived from the real world, directly or indirectly.
Generated	Data that are produced by a model or simulation.
Mean degree	Average degree over all nodes in a network, including any isolates.
Network	A graph where the nodes and edges are intended to represent something, and are not solely an abstract mathematical object.
Node (<i>vertex</i>)	The entities whose relationships are captured by a network.
Isolate	A node with degree of zero.
Structural Edges (<i>prohibited edges, required edges, structural zeros</i>)	A structural edge is an edge that either cannot exist (often called a 'structural zero'), or must exist, <i>in principle</i> . For example, in a network of students where relationships cannot form across classrooms, the absence of a relationship between two students in different classrooms can be represented by a structural zero in an adjacency matrix.